

UAVX Failsafe Operation

Modern receivers have failsafe capability where the channel values go to a known state if the transmitter signal is lost. It is IMPORTANT to set the failsafe values to something sensible e.g. RTH and not pitch/roll/yaw inputs. UAVX will attempt to do what the receiver commands.

Failsafe Detection Paths

There are two independent paths that trigger failsafe behaviour, both handled in `MapRC()`:

1. **SBus failsafe flag:** `sbusDecode()` sets `RCFailsafe` directly from the SBus frame's failsafe bit. `MapRC()` then loads the failsafe presets into `RCInp[]`.
2. **Timeout:** `GenerateFailsafePacket()` detects sustained loss of valid packets and loads the failsafe presets into `RCInp[]` while also setting `RCFailsafe`.

Both paths converge to the same behaviour.

Failsafe Presets

When failsafe is triggered, channel values are set to:

Channel	Value	Effect
NavQual (Ch6)	1650 μ s (65%)	Arm + AltHold enabled
NavMode (Ch5)	2000 μ s	RTH selected
Roll/Pitch/Yaw	1500 μ s	Neutral sticks
Throttle	1050 μ s	Idle

Altitude Hold and Angle Mode Override

At the end of the NavQual processing block in `MapRC()`, if `RCFailsafe` or `RCSignalLost` is set, the code forces: - **Altitude hold enabled** — overrides any other AltHold state - **Angle mode** — overrides Horizon/Rate mode selections

This ensures the aircraft holds altitude and levels attitude regardless of the switch positions that were active before signal loss.

PassThru Override

If `F.PassThru` is active, it overrides all failsafe behaviour: - No nav or altitude hold actions - Pilot-in-command (PIC) mode - Direct stick-to-servo passthrough (mainly used for fixed wing) - `F.PassThru` takes precedence over `RCFailsafe` and `RCSignalLost`